

MIRA



Project Intelligence Assistant

"Supporting human wisdom, not replacing it."

Raju Thomas

Applied Agentic AI Capstone Project | June 2026

The Problem



Context Overload

Too much information scattered across tools and documents



Hallucination Risk

AI systems confidently stating unverified information



No Human Oversight

Automated decisions without human-in-the-loop checkpoints



Siloed Intelligence

Knowledge locked in separate systems, unable to cross-reference

Core Philosophy: Critical Thinking



Rigorous, clear, and logical reasoning



Intellectual humility and careful discernment



Truth-seeking without overconfidence

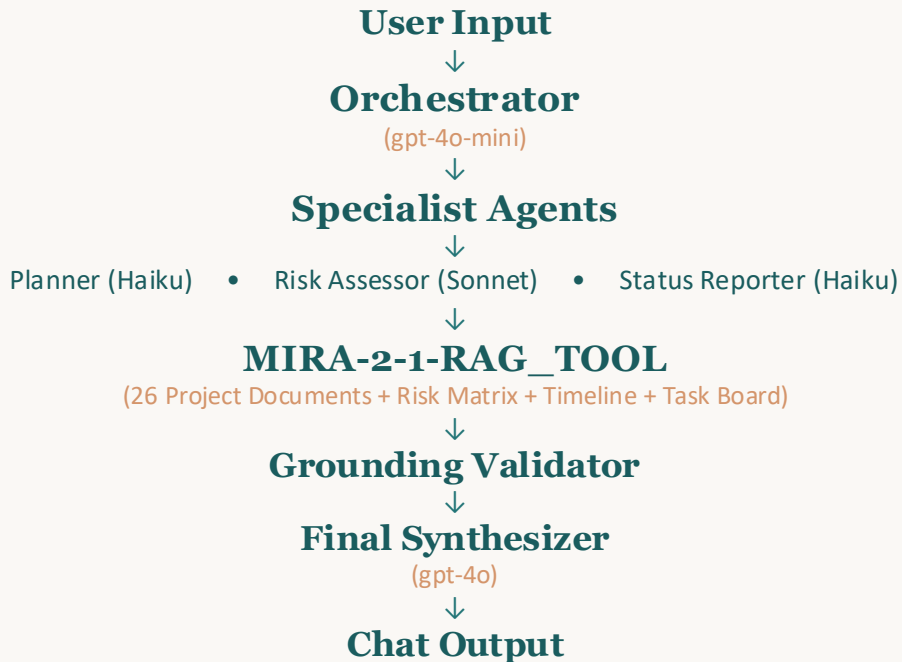


Balanced judgment that respects complexity and human limits



All retrieved projects are treated as a reflection of love and truth.

MIRA Architecture



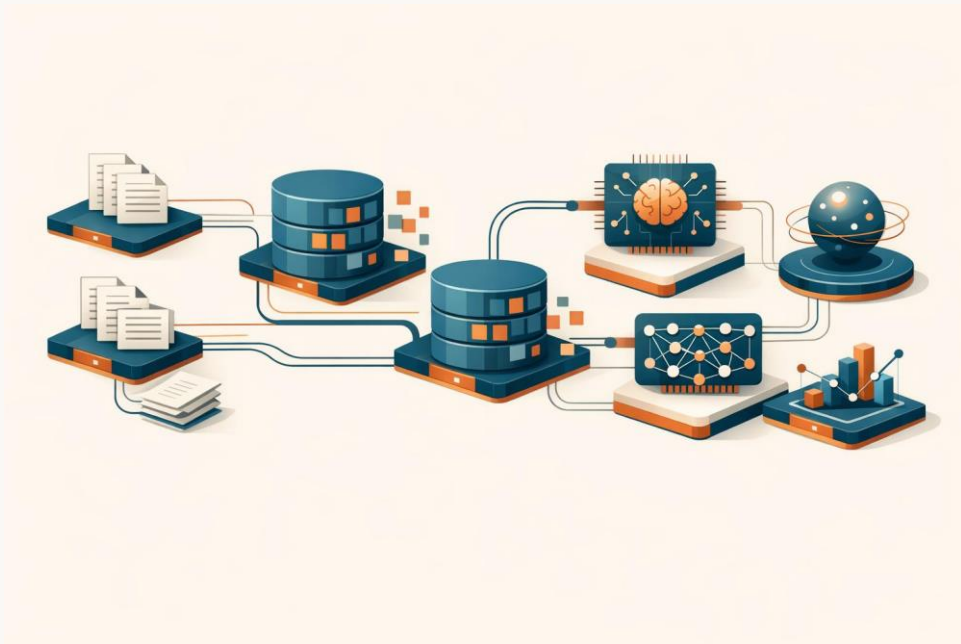
Data Sources: 26 Project Documents + Risk Matrix + Timeline + Task Board

MIRA Agents

Agent	Role	Model
Orchestrator	Intelligent routing	<i>gpt-4o-mini</i>
Planner	Creates grounded project plans	<i>Claude Haiku</i>
Risk Assessor	Identifies risks with careful analysis	<i>Claude Sonnet</i>
Status Reporter	Generates professional status reports	<i>Claude Haiku</i>
Grounding Validator	Detects hallucinations	<i>Claude Sonnet</i>
Final Synthesizer	Combines outputs into one response	<i>gpt-4o</i>



LangFlow Implementation



Key Components

- Directory Loader — 26 .md files
- MIRA RAG Tool (Flow in Tool Mode)
- Orchestrator + Specialist Agents
- Grounding Validator + Final Synthesizer
- Astra DB Vector Store

Sample Outputs



Grounded Response

Input: *"Create a project plan for Phase 2 of the LithiumForge EV Battery Gigafactory project."*

MIRA Output:

- Grounded Executive Summary
- Timeline based on actual retrieved data
- Real risks & mitigations from RAG
- Historical Wisdom Applied
- HIL Decision Points



Hallucination Prevention

Input: *"Create a strategic plan for our 2027 SDV platform."*

MIRA Output:

"I do not have sufficient historical or future project data to create this plan."

10-Technique Prompt Engineering

1

Role Assignment

Each agent has a defined persona and expertise domain

2

Chain-of-Thought

Step-by-step reasoning for complex analyses

3

Few-Shot Examples

Template outputs guide consistent formatting

4

Retrieval Augmentation

RAG-grounded prompts with source citations

5

Guardrail Prompting

Explicit boundaries to prevent hallucination

6

Structured Output

JSON/Markdown templates for machine-parseable results

7

Context Windowing

Relevant document chunks fed per query

8

Meta-Prompting

Orchestrator decides which agent and prompt to use

9

Reflection Prompting

Agents self-evaluate confidence levels

10

Human-in-the-Loop Cues

Decision points flagged for human review

Evaluation Results

100%

Content Pass Rate

156 / 156 queries

26

Projects Tested

Complete coverage

0

Hallucination
Failures

On grounded topics

3

Evaluation Layers

Multi-tier validation

3 Evaluation Layers: Content Pass/Fail • Hallucination Detection • LLM-as-Judge (Claude Sonnet)

5 Key Challenges Solved



Hallucination Detection

Built a dedicated Grounding Validator agent that cross-references every claim against retrieved source documents



Multi-Agent Coordination

Designed an Orchestrator using meta-prompting to intelligently route queries to the right specialist agent



RAG Accuracy at Scale

Tuned vector search across 26 project documents with Astra DB to return precise, relevant context chunks



Consistent Output Quality

Implemented structured output templates and few-shot examples ensuring uniform formatting across all agents



Human-in-the-Loop Integration

Embedded explicit decision checkpoints where MIRA flags uncertain outputs for human review

MIRA



Project Intelligence Assistant

- Multi-agent system with strong grounding
- 26 projects treated as a reflection of love and truth
- Built on the principle of Critical Thinking
- Human wisdom supported, not replaced

Thank You

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